

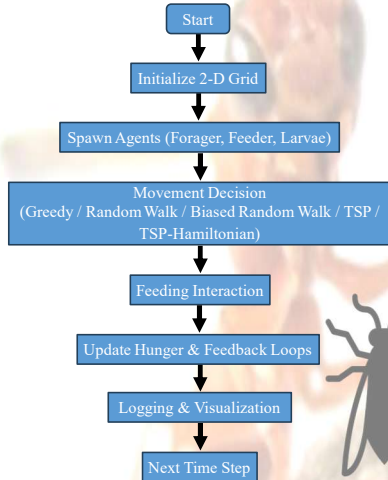
Simulation of Larval Feeding Routes in Social Wasps

Samuel Cornejo, Thaewa Tansarn

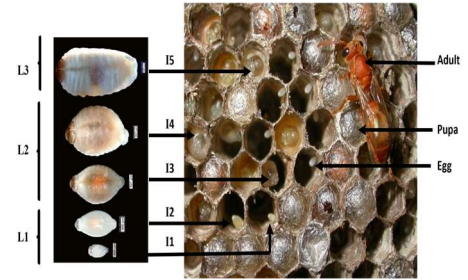
College of Information Science, University of Arizona
Advisor: Dr. Nitika Sharma

INTRODUCTION

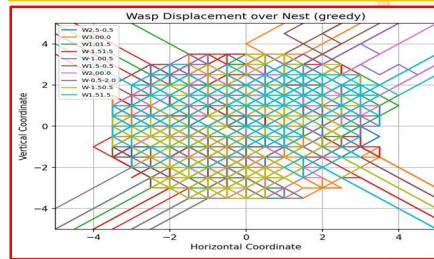
An agent-based simulation of larval feeding in primitively eusocial wasps is constructed using different movement rules (Random walk, biased random, nearest next (greedy), Traveling Salesman Problem (TSP), Hamiltonian path) and feeding mechanisms. These modes illustrate how individual decisions generate colony-level collective feeding patterns.



Courtesy: Thresiamma Varghese

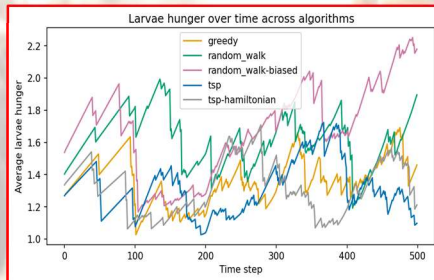


RESULTS



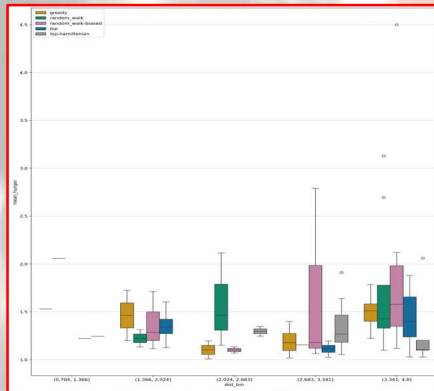
Wasp Trajectories over Simulated Nest.

The movement of each wasp is displayed with different color. Feeder wasps only move inside the nest and foragers outside of it.



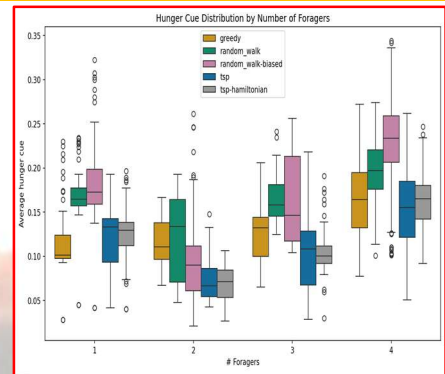
Mean larval hunger across timesteps.

This graph illustrates the wave-like behavior of the average hunger perceived by the larvae over the nest per path-finding mechanism. As it can be seen, the TSP and Greedy pathfinding outmatch the random walks. That is, they maintain the average larvae hunger lower across the simulation timesteps.



Mean larval hunger across distance bins for all movement modes.

Indicating that larvae in outer regions receive less consistent feeding – consistent with empirical observations. The caveat of this observation is that larvae in the center point between the end of the nest and the center of the nest achieve the lowest hunger levels across pathfinding modes. This is consistent with the geometrical mean over the positions of all the wasps in the nest. Note as well that the random walk mechanisms achieve the highest variance from the middle of the nest to the periphery.



Forager recruitment based on local hunger cues

From the graph, it is seen that higher forage counts grow with the local hunger cues. The higher hunger cue related to one forager is tied to the initialization of the simulation where a steady-state regime has not been reached yet.

CONCLUSIONS

The results show that movement strategy may influence feeding performance in the nest. Directed strategies such as the Random-walk and the TSP-based routes maintain lower larval hunger across distance bins and provide more reliable spatial coverage. Larvae located closer to the nest center benefit more from these structured movement patterns.

Gradient-based greedy strategies display higher variability and reduced feeding success, especially at larger distances. This highlights the importance of movement organization when delivering resources in complex environments.

Overall, the simulation demonstrates how local behavioral rules shape colony-level collective behavioral outcomes and provides a practical baseline for future biological modeling or data-driven extensions.

REFERENCES

- [1] N. Sharma and R. Gadagkar, "Heuristic route optimization in a social wasp reflects decentralized solutions to the traveling salesman problem," unpublished manuscript, 2025.
- [2] E. Bonabeau, G. Theraulaz, J. L. Deneubourg, S. Aron, and S. Camazine, "Self-organization in social insects," Trends in Ecology & Evolution, vol. 12, no. 5, pp. 188–193, 1997. doi: 10.1016/S0169-5347(97)01048-3.
- [3] N. Paago, W. Zheng, and P. Nonacs, "Ant foraging: optimizing self-organization as a solution to a traveling salesman problem," Oecologia, vol. 207, art. no. 73, 2025, doi: 10.1007/s00442-025-05720-5.

ACKNOWLEDGEMENTS AND CONTACT

Acknowledgements:

This work was completed under the supervision of Dr. Nitika Sharma, College of Information Science, University of Arizona.

Contact:

samuelcornejo@arizona.edu

ttansarn@arizona.edu

Repository:

github.com/Thaewa/agent-simulator



OBJECTIVES

1. Model larval feeding interactions using agent-based simulation.
2. Implement five baseline pathfinding modes.
3. Compare how each mode affects feeding efficiency, hunger dynamics, and spatial coverage as compared to empirical data.

MATERIALS & METHODS

Simulation Framework

A custom Python-based multi-agent simulator models feeding interactions inside a social wasp nest. The environment is represented as a 2-D grid containing fixed larval cells and walkable paths. All agents operate in discrete time steps and use only local information, consistent with decentralized nest behavior.

Feedback Loops

- ❑ **Larval hunger cues:** Hungry larvae emit stronger fields that attract nearby wasps.
- ❑ **Feeder SURGE behavior:** Feeders without food move toward foragers to obtain food.
- ❑ **Local food transfer:** Wasps feed larvae or nearby wasps based on need.
- ❑ **Dynamic Recruitment:** Wasps will change their roles given the needs of the system.

Pathfinding Modes

- ❑ **Greedy:** Move by following the sign of the local larval-hunger gradient.
- ❑ **Random walk (Biased):** Uniform random movement in the Moore neighborhood. Biased avoids stepping back to previous cell.
- ❑ **TSP (Hamiltonian) route:** Approximate multi-larva visitation using a TSP solver. Hamiltonian makes a cyclic version of the TSP route.